Unit 8 Pre-Test

Coordinate Geometry

Standards assessed: NC.M1.G-GPE.4 · G-GPE.5 · G-GPE.6

Name: _____ **Date:** _____ **Period:** _____

SCORE *3.0* **Basic Understanding**

NC.M1.G-GPE.4 · NC.M1.G-GPE.5 · NC.M1.G-GPE.6 Find midpoints, slopes, and lengths on the coordinate plane.

• *I can find a midpoint, a slope, a distance, and the slopes of parallel and perpendicular lines.*

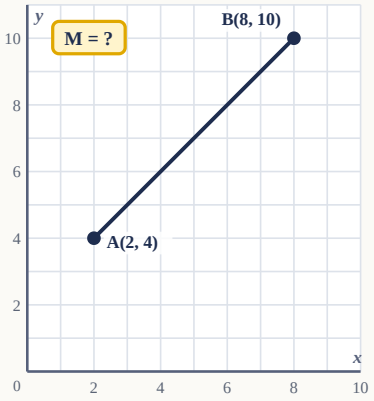
Answer each question. Use the picture to help you.

How to type your answers

point `(5, 7)` fraction `-3/2` root `sqrt(20)` equation `y=2x-2`

1. *Midpoint*

Find the midpoint M of A(2, 4) and B(8, 10).

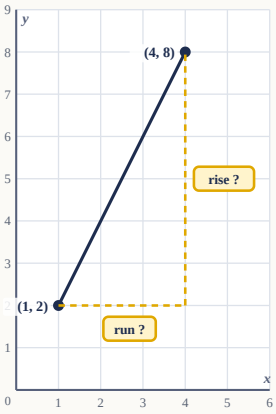


M = _____

answer: (5, 7)

2. *Slope*

Find the slope of the line through (1, 2) and (4, 8).

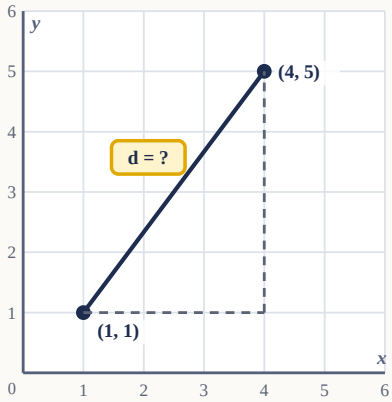


m = _____

answer: 2

3. *Distance*

Find the distance between (1, 1) and (4, 5).

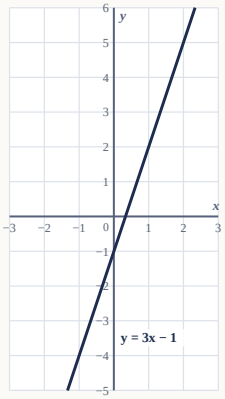


d = _____

answer: 5

4. *Parallel slope*

A line is parallel to y = 3x − 1. What is its slope?

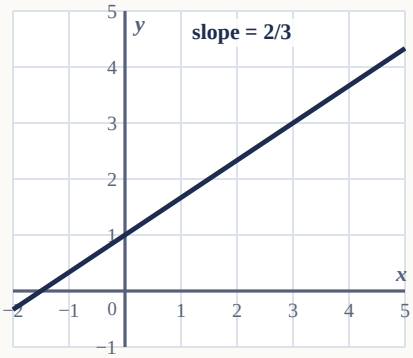


m = _____

answer: 3

5. *Perpendicular slope*

A line has slope $\frac{2}{3}$. What is the slope of a line perpendicular to it?

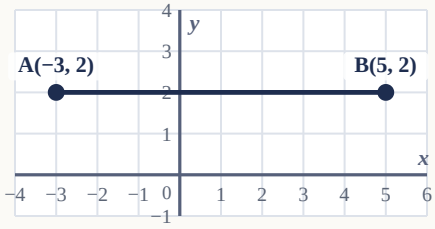


m = _____

answer: $-\frac{3}{2}$

6. *Length*

Find the length of segment AB with A(−3, 2) and B(5, 2).



AB = _____

answer: 8

SCORE 4.0 Target Content

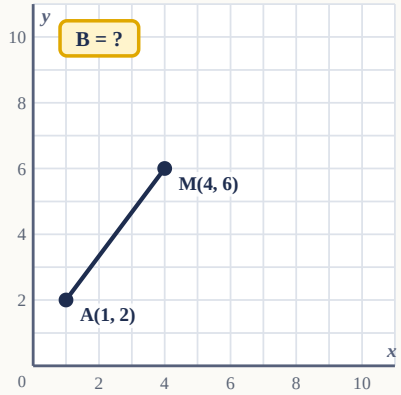
NC.M1.G-GPE.4 · NC.M1.G-GPE.5 · NC.M1.G-GPE.6 Partition segments, write parallel and perpendicular lines, and find perimeter.

• *I can find an endpoint, partition a segment, write equations of parallel and perpendicular lines, and find a perimeter.*

Solve each problem. Show your work on paper.

7. *Endpoint*

M(4, 6) is the midpoint of AB. A is (1, 2). Find B.

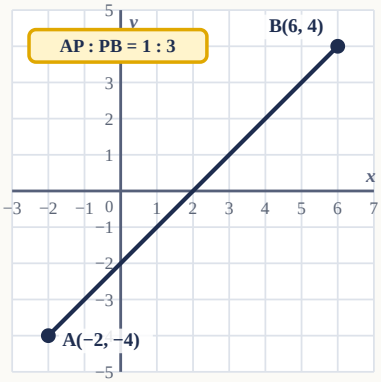


B = _____

answer: (7, 10)

8. *Partition*

Point P is on AB so that AP : PB = 1 : 3. A is (−2, −4) and B is (6, 4). Find P.

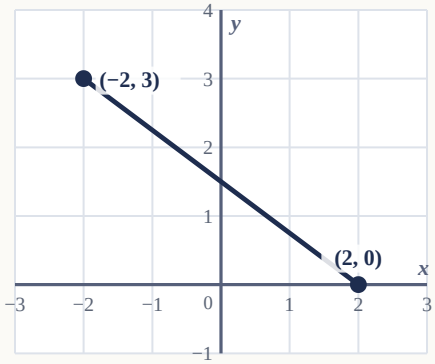


P = _____

answer: (0, −2)

9. *Distance*

Find the distance between (−2, 3) and (2, 0).

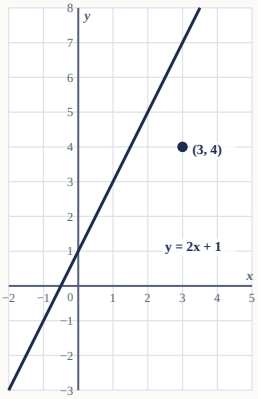


d = _____

answer: 5

10. *Parallel line*

Write the equation of the line parallel to y = 2x + 1 that passes through (3, 4).

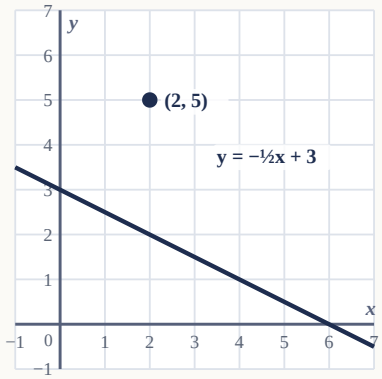


y = _____

answer: y = 2x − 2

11. *Perpendicular line*

Write the equation of the line perpendicular to $y = -\frac{1}{2}x + 3$ that passes through (2, 5).

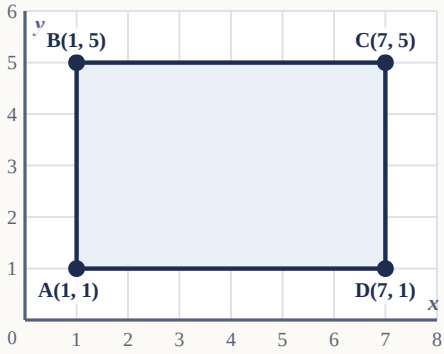


y =

answer: y = 2x + 1

12. *Perimeter*

Find the perimeter of rectangle ABCD.



P =

answer: 20

SCORE 5.0 Advanced Content

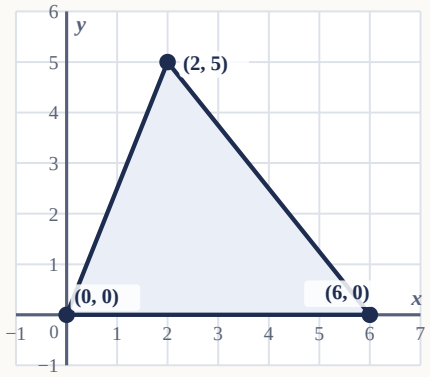
NC.M1.G-GPE.4 · NC.M1.G-GPE.5 · NC.M1.G-GPE.6 Use coordinates to prove facts about figures.

• *I can find area, classify triangles, prove a right angle with slopes, compare lines, and partition in any ratio.*

Solve each problem. Some answers are words.

13. *Area*

Find the area of the triangle with vertices (0, 0), (6, 0), and (2, 5).

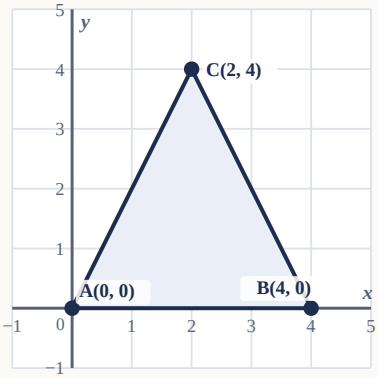


Area = _____

answer: 15

14. *Classify*

Classify triangle ABC by its sides: scalene, isosceles, or equilateral?

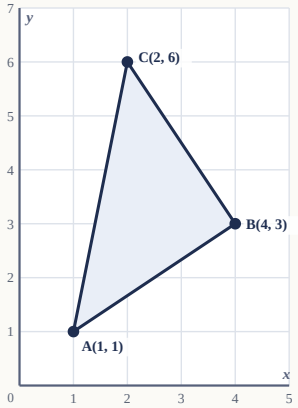


Answer: _____

answer: isosceles (AC = BC = √20)

15. *Right triangle?*

Is triangle ABC a right triangle? Write yes or no.

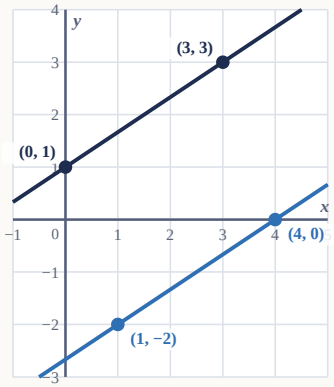


Answer: _____

answer: yes (slopes of AB and BC are $\frac{2}{3}$ and $-\frac{3}{2}$)

16. *Compare lines*

One line goes through (0, 1) and (3, 3). Another goes through (1, −2) and (4, 0). Are they parallel, perpendicular, or neither?

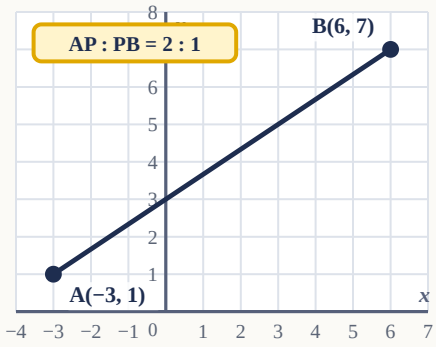


Answer: _____

answer: parallel (both slopes are $\frac{2}{3}$)

17. *Partition*

P is on AB so that AP : PB = 2 : 1. A is (−3, 1) and B is (6, 7). Find P.

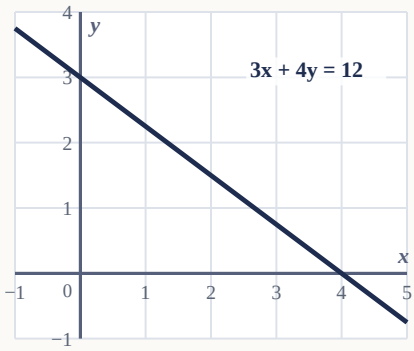


P = _____

answer: (3, 5)

18. *Perpendicular slope*

What is the slope of a line perpendicular to 3x + 4y = 12?



m = _____

answer: $\frac{4}{3}$